Stat 145 (Multivariate Statistics)

Lesson 2.2 Inference on Two Mean Vectors

2025-09-08

Learning Outcomes

1. Derive the statistics for testing hypotheses on two mean vectors.
2. Test hypotheses on two mean vectors.

1. Independent Samples

Let $y_{11}, y_{12}, \cdots, y_{1n_1} \overset{iid}{\sim} N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $y_{21}, y_{22}, \cdots, y_{2n_2} \overset{iid}{\sim} N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$ and $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \boldsymbol{\Sigma}$ is unknown.

We want to test $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ vs $H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$.

It can be shown that $\bar{\boldsymbol{y}}_1 - \bar{\boldsymbol{y}}_2 \sim N_p\left(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2, \left(\frac{1}{n_1+n_2}\right)\boldsymbol{\Sigma}\right)$, where $\boldsymbol{\Sigma}$ is estimated by the pooled sample covariance matrix given by

$$\boldsymbol{S}_p = \frac{(n_1-1)\boldsymbol{S}_1 + (n_2-1)\boldsymbol{S}_2}{n_1+n_2-2}$$

The Hotelling's statistic for testing the above hypothesis is given by

$$T^2 = \frac{n_1 n_2}{n_1+n_2}(\bar{\boldsymbol{y}}_1 - \bar{\boldsymbol{y}}_2)'\boldsymbol{S}_p^{-1}(\bar{\boldsymbol{y}}_1 - \bar{\boldsymbol{y}}_2).$$

Under the null hypothesis, $T^2 \sim T^2_{p,n_1+n_2-2}$, and we reject H_0 if $T^2 \geq T^2_{\alpha,p,n_1+n_2-2}$.

Alternatively, we reject H_0 if

$$\frac{n_1+n_2-p-1}{(n_1+n_2-2)p}T^2 \geq F_{\alpha,p,n_1+n_2-p-1}$$

Example 2.2.1

Four psychological tests were given to 32 men and 32 women. The data is stored in the Excel file *Example 2.2.1.xlsx* in the GitHub repo. The variables are:

- y_1 = pictorial inconsistencies
- y_2 = paper form board
- y_3 = tool recognition
- y_4 = vocabulary

We wish to test:

$$H_0 : \boldsymbol{\mu}_M = \boldsymbol{\mu}_F \text{ vs } H_1 : \boldsymbol{\mu}_M \neq \boldsymbol{\mu}_F$$

Solution:

The sample mean vectors and the sample variance-covariance matrices of male and female participants are shown below:

$$\bar{\boldsymbol{y}}_M = \begin{bmatrix} 15.96875 \\ 15.90625 \\ 27.18750 \\ 22.75000 \end{bmatrix},$$

$$\bar{\boldsymbol{y}}_F = \begin{bmatrix} 12.34375 \\ 13.90625 \\ 16.65625 \\ 21.93750 \end{bmatrix},$$

$$\boldsymbol{S}_M = \begin{bmatrix} 5.192540 & 4.545363 & 6.522177 & 5.250000 \\ 4.545363 & 13.184476 & 6.760081 & 6.266129 \\ 6.522177 & 6.760081 & 28.673387 & 14.467742 \\ 5.250000 & 6.266129 & 14.467742 & 16.645161 \end{bmatrix},$$

and

$$\boldsymbol{S}_F = \begin{bmatrix} 9.136089 & 7.549395 & 4.863911 & 4.151210 \\ 7.549395 & 18.603831 & 10.224798 & 5.445565 \\ 4.863911 & 10.224798 & 30.039315 & 13.493952 \\ 4.151210 & 5.445565 & 13.493952 & 27.995968 \end{bmatrix}.$$

Assuming $\boldsymbol{\Sigma}_M = \boldsymbol{\Sigma}_F = \boldsymbol{\Sigma}$, then the estimate of this common variance covariance matrix is given by

$$\begin{aligned} \boldsymbol{S}_p &= \frac{(n_1 - 1)\boldsymbol{S}_M + (n_2 - 1)\boldsymbol{S}_F}{n_1 + n_2 - 2} \\ &= \begin{bmatrix} 7.164315 & 6.047379 & 5.693044 & 4.700605 \\ 6.047379 & 15.894153 & 8.492440 & 5.855847 \\ 5.693044 & 8.492440 & 29.356351 & 13.980847 \\ 4.700605 & 5.855847 & 13.980847 & 22.320565 \end{bmatrix} \end{aligned}$$

Thus,

$$\begin{aligned} T^2 &= \frac{n_1 n_2}{n_1 + n_2} (\bar{\boldsymbol{y}}_1 - \bar{\boldsymbol{y}}_2)' \boldsymbol{S}_p^{-1} (\bar{\boldsymbol{y}}_1 - \bar{\boldsymbol{y}}_2) \\ &= 97.6015 \end{aligned}$$

and

$$F = \frac{n_1 + n_2 - p - 1}{(n_1 + n_2 - 2)p} T^2 = 23.21971$$

with critical value 2.527907 and p-value $\approx 1.46 \times 10^{-11} < 0.01$. Hence, we reject the null hypothesis and conclude that men and women significantly differ in terms of their scores in the four tests.

REMARK:

When $\mathbf{\Sigma}_M \neq \mathbf{\Sigma}_F$, then we use the following modified Hotelling's T^2 statistic:

$$T^2 = (\bar{\boldsymbol{y}}_1 - \bar{\boldsymbol{y}}_2)' \boldsymbol{S}_T^{-1} (\bar{\boldsymbol{y}}_1 - \bar{\boldsymbol{y}}_2)$$

where

$$\boldsymbol{S}_T = \frac{1}{n_1}\boldsymbol{S}_1 + \frac{1}{n_2}\boldsymbol{S}_2.$$

We can also transform this T^2 statistic into an F statistic as follows

$$F = \frac{n_1 + n_2 - p - 1}{(n_1 + n_2 - 2)p} T^2 \sim F_{p,v}$$

where

$$\frac{1}{v} = \sum_{i=1}^{2} \frac{1}{n_i - 1} \left\{ \frac{(\bar{\boldsymbol{y}}_1 - \bar{\boldsymbol{y}}_2)' \boldsymbol{S}_T^{-1} (\frac{1}{n_i}\boldsymbol{S}_i) \boldsymbol{S}_T^{-1} (\bar{\boldsymbol{y}}_1 - \bar{\boldsymbol{y}}_2)}{T^2} \right\}^2$$

We will reject H_0 if $F \geq F_{\alpha,p,v}$.

Example 2.2.2

Consider the data in *Example 2.2.1* and test $H_0 : \boldsymbol{\mu}_M = \boldsymbol{\mu}_F$ vs $H_1 : \boldsymbol{\mu}_M \neq \boldsymbol{\mu}_F$ assuming $\mathbf{\Sigma}_M \neq \mathbf{\Sigma}_F$.

Solution: Let as a classroom exercise!

2. Paired or Dependent Samples

Consider two random samples $\mathbf{y}_1, \mathbf{y}_2, \cdots, \mathbf{y}_n$ and $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n$, where there is a pairing between the i^{th} observations $\mathbf{y}_i$ and $\mathbf{x}_i$, $\forall i, i = 1, 2, \cdots, n$. To test

$$H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2 \text{ vs } H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$$

we must assume that

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{x} \end{bmatrix} \sim N_{p+p}\left(\begin{bmatrix} \boldsymbol{\mu}_y \\ \boldsymbol{\mu}_x \end{bmatrix}, \begin{bmatrix} \mathbf{\Sigma}_y & \mathbf{\Sigma}_{yx} \\ \mathbf{\Sigma}_{xy} & \mathbf{\Sigma}_x \end{bmatrix} \right)$$

then use

$$\mathbf{d}_i = \mathbf{y}_i - \mathbf{x}_i \sim N_p(\boldsymbol{\mu}_y - \boldsymbol{\mu}_x, \mathbf{\Sigma}_d)$$

where

$$\mathbf{\Sigma}_d = \mathbf{\Sigma}_y + \mathbf{\Sigma}_x - \mathbf{\Sigma}_{yx} - \mathbf{\Sigma}_{xy}.$$

The test statistic is

$$T^2 = n\bar{\boldsymbol{d}}' \boldsymbol{S}_d^{-1} \bar{\boldsymbol{d}}$$

where $\bar{\boldsymbol{d}}$ and $\boldsymbol{S}_d$ are the sample mean and covariance of the sample pairwise differences.

If H_0 is true, $T^2 \sim T^2_{p,n-1}$, the Hotellings T^2 distribution with parameters p and $v = n-1$. We reject H_0 if $T^2 \geq T^2_{\alpha,p,v}$.

An alternative test statistic is

$$F = \frac{n-p}{(n-1)p} T^2$$

which follows a $F_{p,n-p}$ distribution. Consequently we reject H_0 if $F \geq F\alpha, p, v$.

Example 2.2.3

Municipal wastewater treatment plants are required by law to monitor their discharges into rivers and streams on a regular basis. Concern about the reliability of data from one of these self-monitoring programs led to a study in which samples of effluent were divided and sent to two laboratories for testing. One-half of each sample was sent to the Central Analytical Services Laboratory (CASL), and one-half was sent to a private commercial laboratory routinely used in the monitoring program. Measurements of biochemical oxygen demand (BOD) and suspended solids (SS) were obtained, for $n = 11$ sample splits, from the two laboratories. Let $y_1 = BOD(Private), y_2 = SS(Private), x_1 = BOD(CASL),$ and $x_2 = SS(CASL)$.

Sample	y_1	y_2	x_1	x_2
1	6	27	25	15
2	6	23	28	13
3	18	64	36	22
4	8	44	35	29
5	11	30	15	31
6	34	75	44	64
7	28	26	42	30
8	71	124	54	64
9	43	54	34	56
10	33	30	29	20
11	20	14	39	21

Do the two laboratories' chemical analyses agree?

Solution:

We wish to test

$$H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2 \text{ vs } H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$$

where $\boldsymbol{\mu}_1$ be the mean vector of BOD and SS for Lab 1, and $\boldsymbol{\mu}_2$ be the mean vector of BOD and SS for Lab 2.

You can verify that the pairwise differences for BOD and SS are

bod	ss
-19	12
-22	10
-18	42
-27	15
-4	-1
-10	11
-14	-4
17	60
9	-2
4	10
-19	-7

Consequently, the mean vector $\bar{\boldsymbol{d}}$ and the covariance matrix $\boldsymbol{S}_d$ are, respectively

$$\bar{\boldsymbol{d}} = \begin{bmatrix} -9.36 \\ 13.27 \end{bmatrix}$$

and

$$\boldsymbol{S}_d = \begin{bmatrix} 199.25 & 88.31 \\ 88.31 & 418.62 \end{bmatrix}.$$

Thus, the test statistic is

$$\begin{aligned} T^2 &= n\bar{\boldsymbol{d}}'\boldsymbol{S}_d^{-1}\bar{\boldsymbol{d}} \\ &= 11 \begin{bmatrix} -9.36 \\ 13.27 \end{bmatrix}' \begin{bmatrix} 199.25 & 88.31 \\ 88.31 & 418.62 \end{bmatrix}^{-1} \begin{bmatrix} -9.36 \\ 13.27 \end{bmatrix} \\ &\approx 13.63 \end{aligned}$$

which when transformed to F becomes

$$\begin{aligned} F &= \frac{n-p}{(n-1)p} T^2 \\ &= \left[\frac{11-2}{(11-1)\times 2}\right] 13.63 \\ &= 6.13 \end{aligned}$$

The critical value of the F distribution at the 5% level of significance is 4.26. In addition, the associated p-value of the calculated F statistic is 0.0209.

Therefore, we reject H_0 and conclude that there is a nonzero mean difference between the measurements of the two laboratories. It appears, from inspection of the data, that the private lab tends to produce lower BOD measurements and higher SS measurements than the CASL.